Hang Mixed Garments on a Garment Rack SOP (1x Episode: 4 Garments)

One episode hangs four mixed garments on the garment rack: one **T-shirt**, one **button shirt**, one pair of **trousers**, and one pair of **shorts**. The T-shirt and the button shirt are **tops**. The trousers and the shorts are **bottoms**. Each garment is lifted off the garment pile with the left gripper and held hanging in the air at the center of the table. The right gripper takes one hanger and puts it on the garment there. Then the right gripper carries the garment to the rail and sets the hook on the rail.

A top gets the hanger down through its **neck opening**. The left gripper pushes the left shoulder onto the left side of the hanger, and the right gripper puts the right shoulder onto the right side. A bottom hangs folded over the **bar** of the hanger. The right gripper slides the bar into the **middle fold** from the open end, and the left gripper lays the fold on the bar.

The order never changes: lift and hold, take the hanger, put the hanger on, hang. One garment is off the pile at a time. No garment is left on the rail with a shoulder off its hanger end, with its fold off the middle of the bar, or with cloth folded the wrong way. If a shoulder slips off its hanger end or a fold slides along the bar, either gripper may set it right at any time. That is a fix, not a step.

Nothing is laid on the tabletop. Every garment goes from the pile into the air, from the air onto the hanger, and from the hanger onto the rail. The rack never moves.

The left gripper takes every garment off the pile and holds it at the fit spot while the hanger goes on. The right gripper does the main work: it takes every hanger, puts it on the garment, and carries the garment to the rail and hangs it. The left gripper supports: it holds the garment for the fit, pushes the left shoulder of a top onto the hanger, and lays the fold of a bottom on the bar.

The table is set up in one of three ways. Only the garment pile moves; the rack, the fit spot, and the hanger pile are in the same place in all three. The garment pile always sits where the left arm reaches it.

- **Config L1:** the garment pile is in the back-left corner, left of the rack's left upright.
- **Config L2:** the garment pile is in the front-left corner.
- **Config M:** the garment pile is at the front-center, on the center line at the front edge.

Where a step depends on the setup it says so on an **IF** line: look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the garment pile starts at the back-left (**Config L1**), the front-left (**Config L2**) or the front-center (**Config M**). One config per episode, chosen before recording and never changed mid-episode.
- **Left-arm rule:** the left gripper takes every garment off the pile in every config. The pile is never staged on the right, and no arm reaches across the table for a garment.
- **Fixed roles:** everything else is the same in all three configs. The right gripper takes every hanger from the hanger pile at the front right, puts it on the garment, and hangs the garment. The left gripper holds the garment for the fit and puts on the left shoulder of a top or the fold of a bottom.
- **Mixed order:** the four garments lie in the pile in a mixed order that changes from episode to episode. The order is not part of the config. Each garment is worked by its kind, top or bottom, and hung on the rail in the order it comes off the pile.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the garment rack across the back, the garment pile in the start zone for this episode's config, the hanger pile at the front right, and the clear center of the table where each garment is held.
3. The rail is in frame end to end, from its **left end** to its **right end**, with clear space above it and below it, so every hook can be seen landing and every hanging garment can be seen from its hook to its lowest point.
4. The **fit spot** is in frame from the top of a held garment down to its lowest point.
5. Both arms are at home with grippers open.
6. The tabletop holds nothing except the garment rack, the garment pile, the hanger pile, and robot hardware.
7. Nothing blocks the paths between the garment pile, the hanger pile, the fit spot, and the rail.
8. The right arm reaches the whole hanger pile, the fit spot, and the whole rail from its left end to its right end. It does not lean out or stretch to a joint limit to get there.
9. The left arm reaches the whole garment pile in every config, the fit spot, and the whole rail from its left end to its right end. It does not lean out or stretch to a joint limit to get there.

Materials checklist

1. One **garment rack** stands across the back of the table, in the middle of the table's back edge. Its **left upright** stands at the back left of center and its **right upright** at the back right of center, so the **rail** runs left to right across the back.
2. The rack stands firm on its own base. It does not rock, slide, or lift when a hanger is set on the rail, and no gripper ever holds it up.
3. The rail is bare, straight, and clear along its whole length. It is high enough that every garment on a hanger hangs clear of the tabletop and the garment pile, and low enough that both grippers reach the top of every hanging garment.
4. The rail is long enough that all four hangers hang on it in **one run**. Garments may touch each other on the rail.
5. Four garments sit in one **garment pile** in the start zone for this episode's config, in a mixed order, each laid straight on the one below it:
   - **Config L1:** back-left corner, left of the rack's left upright
   - **Config L2:** front-left corner
   - **Config M:** front-center, on the center line at the front edge
6. The four garments are one T-shirt, one button shirt, one pair of trousers, and one pair of shorts.
7. Each top lies **face up** and **folded in half**, collar toward the far side, with both sleeves stretched flat. The button shirt has every button closed except the **top button**, which is open.
8. Each bottom lies **legs together** and folded once across the middle of its legs, with the **middle fold** toward the front edge and the waistband and hems toward the far side.
9. Every top's neck opening opens wide enough, once the top hangs, that a hanger passes down through it with clear space around the hanger. A top whose neck opening will not open that far is not staged.
10. Every garment is dry, clean, and pressed, with no tears and no open seams.
11. Four **hangers** lie flat in one **hanger pile** at the front right, each laid straight on the one below it. Every **hook** points toward the far side with its **hook opening** facing the back. The **open end** of every bar is on the left.
12. All four hangers are the same shape and size: two sound **shoulder ends**, a hook that is not bent, and a straight bar that is open at its left end. A hanger that will not sit straight on the rail is replaced before the episode.
13. Before collection, check the whole load fits: all four garments hanging in one run on the rail, none on an upright and none touching the tabletop.

Workspace layout

- **Garment rack:** across the back of the table, in the middle of the back edge. It holds the **rail**, the **left upright** at the back left of center, and the **right upright** at the back right of center. The rack stays where it is for the whole episode and no gripper ever lifts it.
- **Rail:** the bar the hangers go on. Its **left end** is the end toward the left of the table and its **right end** is the end toward the right. The first garment of the episode goes at the left end and each garment after it goes to the right of the last one.
- **Fit spot:** the air above the center of the table, just in front of the rail, where the left gripper holds each garment while the hanger goes on. Both arms reach it. The tabletop under it stays bare.
- **Garment pile:** the four garments, in the start zone for this episode's config: the back-left corner (**Config L1**), the front-left corner (**Config L2**), or the front-center (**Config M**). The left arm can reach it.
- **Hanger pile:** at the front right, where the right arm can reach. The four hangers lie flat in one stack. The top hanger is always the next one.

The hangers are always staged on the right and the right arm takes every one. Every garment is taken by the left arm and goes straight up to the fit spot at the center, so nothing is carried across the table.

Arm assignments

- **Right gripper:** works. It takes each hanger by the hook from the top of the hanger pile and puts it on the garment at the fit spot. For a top, it pushes the hanger down through the neck opening, lets go of the hook, puts the right shoulder onto the right side of the hanger, and carries the top to the rail by that hold. For a bottom, it slides the bar into the middle fold and carries the bottom to the rail by the hook. It hangs every garment. It never takes a garment off the pile.
- **Left gripper:** supports. It takes each garment off the pile, holds it still at the fit spot while the hanger goes on, pushes the left shoulder of a top onto the hanger, and lays the middle fold of a bottom on the bar. It never takes a hanger.
- Wherever the left gripper holds, the right gripper works clear of it. The two grippers never crowd the same spot.

Vocabulary

- **Garment:** one piece of clothing, hung whole. There are four in an episode and they are worked one at a time.
- **Top:** the T-shirt or the button shirt. A top hangs from the hanger by its two shoulders.
- **Bottom:** the trousers or the shorts. A bottom hangs folded over the bar of the hanger.

- **Garment pile:** the four garments stacked in the start zone: the back-left corner (**Config L1**), the front-left corner (**Config L2**), or the front-center (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Face up:** a top lies with its front showing and its back against the pile, so the left shoulder point is on the left of the pile.
- **Folded in half:** a top lies face up on the pile folded once across its middle, hem folded up toward the collar, collar toward the far side, with both sleeves stretched out flat to the sides. The fold opens by itself when the top is lifted and hangs.
- **Legs together:** a bottom lies with one leg flat on top of the other, both legs lined up, front crease on front crease.
- **Middle fold:** the fold across the middle of a bottom's legs, with the waistband half on one side and the hem half on the other. On the pile it faces the front edge. The **middle of the fold** is the spot halfway along it.
- **Neck opening:** the hole at the top of a held top that the neck goes through, standing open just to the right of the left gripper's pinch. For the button shirt it is the open collar above the top button. A hanger goes into a top through this and nowhere else.
- **Left shoulder point:** the top corner of a top where the left shoulder seam meets the sleeve, on the left as seen from above with the top face up. It is the part of a top the left gripper pinches to lift and hold it. The **right shoulder point** is the matching corner on the right.
- **Held top:** a top hanging from the left gripper's pinch on the left shoulder point, front facing the front edge, the fold fallen open, both sleeves hanging down, nothing caught up over the neck opening, and the hem hanging free and clear of the tabletop and the garment pile.
- **Held bottom:** a bottom hanging from the left gripper's pinch at the middle of the middle fold, the fold flat across the top, legs together, one half hanging in front of the other, and both halves hanging free and clear of the tabletop and the garment pile.
- **Fit spot:** the air above the center of the table, just in front of the rail, where the left gripper holds each garment for the hanger. Both arms reach it.
- **Forced fit:** the hanger pressed against the neck opening so it pulls out of shape or a seam goes tight, or the bar pushed against the cloth so the fold drags along with it. Lifting the hanger back out and bringing it in again is not a forced fit.
- **Hanger:** the frame the garment hangs on. It has two **shoulder ends**, one **hook**, and one **bar** across the bottom.
- **Hanger pile:** the four hangers stacked flat at the front right, hooks toward the far side, hook openings facing the back, and the open end of every bar on the left. The top hanger is always the next one.
- **Hook:** the curved top of the hanger. It is the only part of the hanger a gripper takes from the pile, and the only part that ever sits on the rail. Its **hook opening** is the gap under the curve, and it faces the back on every hanger.

- **Shoulder end:** the tip at either end of the hanger's top. The **left side of the hanger** ends in the left shoulder end and the **right side** in the right shoulder end, as seen from the front edge.
- **Bar:** the straight piece across the bottom of the hanger. Its left end is not joined to the hanger: that is the **open end**, and a bottom slides onto the bar from there. Its right end is joined to the hanger and is the **closed end**.
- **On the bar:** a bottom's middle fold lies across the middle of the bar, flat and level, legs together, one half hanging in front of the bar and one behind it, the two halves the same length.
- **On the hanger (top):** one shoulder end sits inside each shoulder of the top, the hook stands up out of the neck opening, the collar or neckband lies flat around the hook, and nothing is caught up on the hanger.
- **Lift away:** raise the gripper or item straight up until there is daylight under it, before anything moves sideways.
- **Carry level:** move the item across at one height, turned the same way as when it was picked up, without turning, swinging, or shaking it.
- **One run:** all four hangers sit on the rail in a single line, in the order they were hung, left end first, with no garment behind another, no hook on another hanger, and no garment on an upright. Garments in the run may touch each other.
- **Settled:** the item stays put when the gripper lifts away, with nothing rocking, sliding, swinging off, or falling.
- **Let-go point:** a gripper opens only at one of these: the left gripper at the fit spot once both shoulders of a top are on the hanger, or once the fold of a bottom lies on the bar; the right gripper on the hook of a top once the left shoulder is on, or over the rail with the hook already resting on the rail; either gripper on a hanging garment with the cloth resting where it was laid.

Steps

Run Steps 1 to 4 four times, once per garment. Then run Step 5 once. One garment is off the pile at a time, and it hangs settled on the rail before the next one comes off the pile.

Only the carry in Step 1.2 depends on the config: the **left gripper** takes every garment, and the IF line says which way it carries the garment to the fit spot. Every other config line is the same in all three configs. Steps 1.1, 2, 3, and 4 also have one IF line per garment kind: look at the garment on top of the pile and follow the top line or the bottom line.

Any grasp or fit may be retried. If a gripper slips off, a hook slides out of the gripper, or a hanger does not go in clean, the gripper opens, backs off, and tries again. Nothing is ever forced to save a retry.

Setting a shoulder back on its hanger end, or sliding a fold back to the middle of the bar, is a fix, not a step. Do it only if needed, at any time before the next garment comes off the pile, with whichever gripper is free.

Step 1: Lift the next garment and hold it at the fit spot

Goal: the next garment hangs from the left gripper at the fit spot, the right way round and not twisted.

1.1 Pick the garment off the pile

Look at the top garment of the garment pile.

- **IF it is a top:** the **left gripper** pinches the **left shoulder point** of the top garment, then lifts it straight up until the whole top hangs clear of the pile and the fold has fallen open.
- **IF it is a bottom:** the **left gripper** pinches the top garment at the **middle of the middle fold**, then lifts it straight up until the whole bottom hangs clear of the pile.

Take only the top garment. Never pull one out from under another. Never lift a top by a sleeve, the collar, or the hem, and never lift a bottom by the waistband, a leg end, or a corner of the fold. If the pinch slips or misses its spot, the **left gripper** opens and takes it again.

1.2 Carry it to the fit spot

Look where the garment pile is.

- **IF the garment pile is at the back-left (Config L1):** the **left gripper** carries the garment level to the right to the **fit spot**.
- **IF the garment pile is at the front-left (Config L2):** the **left gripper** carries the garment level back and to the right to the **fit spot**.
- **IF the garment pile is at the front-center (Config M):** the **left gripper** carries the garment level toward the far side to the **fit spot**.

Then, in all three:

- The **left gripper** holds the garment still at the fit spot, clear of the tabletop and the garment pile, until Step 3 says it can let go.

Check: a top is a **held top**: front facing the front edge, fold fallen open, both sleeves hanging down, the neck opening standing open, and nothing caught up over it. A bottom is a **held bottom**: fold flat across the top, legs together, not twisted. The pile holds the remaining garments, still neat.

If a top has not fallen open, hangs twisted, or has its back to the front edge, or if a bottom's legs have come apart or it hangs twisted, the **left gripper** puts it back down on the pile, opens, and takes it again.

Step 2: Take the next hanger

Goal: one hanger in the right gripper, held ready beside the held garment.

- The **right gripper** closes on the **hook** of the **top hanger of the hanger pile** and lifts it straight up clear of the pile.
- **IF the garment is a top:** the **right gripper** carries the hanger level to the fit spot and holds it above the **neck opening**, hook up.
- **IF the garment is a bottom:** the **right gripper** carries the hanger level to the right of the held bottom, hook up, the bar level and just below the middle fold, the **open end** pointing at the bottom.
- The **left gripper** holds the garment still the whole time.

Take the top hanger only, by the hook. Never take a hanger by a shoulder end or the bar, never slide one out from under another, and never lift two at once. If the hook slides out of the gripper, the **right gripper** opens and closes on the hook again.

Check: the **right gripper** holds one hanger by the hook, hook up, open end of the bar on the left. The hanger pile holds the remaining hangers, still neat. The held garment has not moved.

Step 3: Put the hanger on the garment

Goal: the garment on the hanger, still at the fit spot, ready to carry.

- **IF the garment is a top:** run 3.1.
- **IF the garment is a bottom:** run 3.2.

3.1 Fit a top

- The **right gripper** pushes the hanger down through the **neck opening**, near the left gripper's pinch, with clear space around the hanger.
- The **right gripper** keeps putting the hanger in until the left side of the hanger is in a good position to enter the left shoulder.
- The **left gripper** pushes that side of the top toward the hanger, so that the left side of the hanger goes into the left shoulder. The **left gripper** keeps its hold on the shoulder, with the hanger inside it.
- The **right gripper** lets go of the hook, takes the **right shoulder point**, brings it over to the right side of the hanger, and puts it on.
- The **right gripper** keeps its hold on the right shoulder.

The hanger goes in through the neck opening and nowhere else. Never take it in through a sleeve or up from the hem. If it does not go in, the **right gripper** lifts it out and brings it down again.

Check: the top is **on the hanger**: one shoulder end in each shoulder, the hook standing up out of the neck opening, the collar or neckband lying flat around the hook and not turned under. The

neck opening is not stretched. The **left gripper** holds the left shoulder and the **right gripper** holds the right shoulder.

If a shoulder end is in a sleeve or outside the top, the gripper on that side lifts that shoulder off and puts it on again. If the collar is turned under, the **right gripper** lifts it flat. If the hook went down inside the top, the **right gripper** takes the hook, lifts the hanger back out through the neck opening, and brings it down again.

3.2 Fit a bottom

- The **right gripper** moves the hanger level to the left, open end first, so the bar slides into the middle fold from the right, just below the left gripper's pinch.
- The **right gripper** keeps sliding until the middle of the bar is under the middle of the fold.
- The **left gripper** brings the fold down until it lies on the bar, then opens and lifts away to the left.
- The **right gripper** keeps its hold on the hook.

The bar goes into the fold from the open end and nowhere else. Never push the bar into a leg opening or the waistband. If the bar catches on the cloth, the **right gripper** slides it back out and brings it in again.

Check: the bottom is **on the bar**: the fold lies across the middle of the bar, flat and level, legs together, one half in front of the bar and one behind it, the two halves the same length. The **right gripper** holds the hook.

If the fold sits off the middle or one half hangs longer than the other, the **left gripper** takes the fold again at its middle, lifts it a little off the bar, and lays it back across the middle.

Step 4: Hang the garment on the rail

Goal: the garment hanging settled on the rail in one run, to the right of the garment already there.

- **IF the garment is a top:** the **left gripper** opens and moves away to the left. The **right gripper** carries the top level to the rail by its hold on the right shoulder, with the hanger inside.
- **IF the garment is a bottom:** the **right gripper** carries the bottom level to the rail by the hook.
- The **right gripper** keeps the garment clear of the garment pile and any garment already hanging.
- The **right gripper** sets the **hook** over the rail, **hook opening** facing the back, until the hook rests on the rail and the garment hangs free. Then it opens and lifts away.
- Put the first garment at the **left end** of the rail. Put each garment after it to the right of the last one. Garments may touch each other; no garment goes behind, on top of, or over the

hook of another.
- The **left gripper** waits clear of the rail until the right gripper is off the garment.

Never carry a garment over one already hanging.

Check: the hook is fully over the rail with its opening facing the back, and the garment hangs free and **settled**. A top still has both shoulders on their hanger ends and its front facing the front edge. A bottom still lies across the middle of the bar with its halves the same length. All the garments hung so far sit in **one run** from the left end, hooks all facing the same way. The rack has not moved.

If the hook is not fully on the rail, or if the garment lands out of the run, the **right gripper** takes the hook, lifts the garment off the rail, and hangs it again. Do not push or slide a hanger along the rail after letting go. If a shoulder has come off its hanger end or a fold has slid along the bar, either gripper sets it right. This is a fix, not a step.

Go back to Step 1 for the next garment. After the fourth garment, go on to Step 5.

Step 5: End the episode

Goal: recording ends with all four garments hanging on the rail.

- Look once along the rail: four garments in one run from the left end, each hanging right on its hanger, hooks facing the back, and both piles bare.
- Return both arms home with grippers open. Homing is the last thing the arms do.
- Stop recording.

Check: both arms are at home, both grippers are open, and neither holds anything.

Expected state: the four garments hang still on the rail, the rack has not moved, and the recording has stopped.

After the episode: reset the workspace

This reset is not recorded.

1. Take each garment off the rail by its hook, one at a time, and take the hanger out: for a top, lift it back out through the neck opening; for a bottom, slide the bar out of the fold toward the open end.
2. Stack the four hangers back into the hanger pile at the front right, flat, hooks toward the far side with the hook opening facing the back, the open end of every bar on the left, each laid straight on the one below.
3. Smooth each garment out and press it if it is creased or carries a mark from the hanger. A garment that still shows a hanger mark is not staged again.

4. Rebuild the garment pile in the start zone for the next episode's config: back-left corner (Config L1), front-left corner (Config L2), or front-center (Config M). Put the four garments in a new mixed order.
5. Lay each top face up and folded in half, hem folded up toward the collar, collar toward the far side, both sleeves stretched flat. Close every button of the button shirt except the top button.
6. Lay each bottom legs together and folded once across the middle of its legs, middle fold toward the front edge.
7. Check every neck opening still opens wide enough for a hanger to pass with clear space around it, and that no neck opening is stretched or wavy.
8. Check all four hangers: no bent hook, no cracked shoulder end, a straight bar with its open end still open, and each one still sits straight on the rail.
9. Check the rack stands firm, the rail is straight and clear along its whole length, and the uprights have not shifted.
10. Replace anything damaged: a garment that is torn, stained, or damp; a top whose neck opening will not open or hold its shape; a bent hanger or a bar that has closed up; or a rack that rocks or will not stand straight.
11. Wipe the tabletop and check the center of the table is bare.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Keep the episode with its violation tags. Do not delete it just because a rule was broken.

Tag a violation only when something is forced, the order is broken, or the episode moves on with a fault left in place.

Violations

Violation: Config misaligned

- **Visible cue:** what the arms do does not match the config on the table: the garment pile is not in the start zone for the config; the right gripper takes a garment off the pile; a gripper reaches across the table for a garment; or the wrong IF line is followed in Step 1.2.
- **SOP rule broken:** Step 1.2, the start position and the left-arm rule (the left gripper takes every garment off the pile in every config, no arm reaches across the table for a garment, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the garment pile is before the first reach, then follow that config's IF line in Step 1.2.

Violation: Wrong order, or two garments worked at once

- **Visible cue:** the hanger comes over before the garment is held at the fit spot; the garment goes to the rail before it is on the hanger; or a second garment or hanger comes off its pile before the last garment hangs settled on the rail.
- **SOP rule broken:** Steps 1 to 4, lift and hold, take the hanger, put the hanger on, then hang, in that order, one garment at a time.
- **Coaching note:** finish the garment on the rail before you touch the next one.

Violation: Wrong kind line followed

- **Visible cue:** a top is fitted the bottom way (slid onto the bar, hung folded over the bar), or a bottom is fitted the top way (hanger pushed down into the waistband or a leg, hung by its waistband on the shoulder ends).
- **SOP rule broken:** Steps 1.1, 2, and 3, look at the garment and follow the top line or the bottom line: a top goes on the hanger through the neck opening by its shoulders, and a bottom goes over the bar by its middle fold.
- **Coaching note:** look at what is on top of the pile first. Tops by the shoulders, bottoms over the bar.

Violation: Wrong pickup from the garment pile

- **Visible cue:** the left gripper takes a garment other than the top one, works one out from under or between the others, slides one off sideways, lifts two garments in one grasp, or shakes or swings a garment to open it.
- **SOP rule broken:** Step 1.1, only the top garment is taken, and it is lifted straight up clear of the pile.
- **Coaching note:** come down on the top one and lift straight up. Nothing gets dug out.

Violation: Item taken by the wrong part

- **Visible cue:** a top is lifted by a sleeve, the collar, or the hem rather than the left shoulder point; a bottom is lifted by the waistband, a leg end, or a corner of the fold rather than the middle of the middle fold; and the episode carries on with that hold. Or a hanger is taken by

a shoulder end or the bar rather than the hook, a hanger other than the top one is taken, or two are lifted at once.

- **SOP rule broken:** Steps 1.1 and 2, a top comes off the pile at the left shoulder point, a bottom at the middle of the middle fold, and every hanger is taken by the hook and always the top one of the pile.
- **Coaching note:** shoulder point for a top, middle of the fold for a bottom, hook for every hanger. Land on the grasp point or open and take it again.

Violation: Garment not held at the fit spot

- **Visible cue:** the left gripper lays the garment on the tabletop, drapes it over the rack or the pile, holds it off to one side instead of at the center in front of the rail, holds it touching the tabletop or the pile, or moves it about while the hanger is coming on.
- **SOP rule broken:** Steps 1.2 to 3, the left gripper holds the garment still at the fit spot, clear of the tabletop and the pile, from the lift until its let-go point in Step 3.
- **Coaching note:** lift it to the middle and hold it still. The garment never touches the table.

Violation: Hanger brought to a garment that is not hanging right

- **Visible cue:** the hanger comes over while a top has not fallen open, hangs twisted, has its back to the front edge, or has a sleeve caught up over the neck opening; or while a bottom hangs twisted or with its legs apart; and the garment is not put back on the pile and taken again first.
- **SOP rule broken:** Step 1.2, a garment that is not hanging right is put back down on the pile and taken again before the hanger comes over.
- **Coaching note:** look at the garment before you reach for the hanger. If it is not hanging right, put it back and take it again.

Violation: Hanger in through the wrong opening

- **Visible cue:** the hanger goes into a top through a sleeve or up from the hem instead of down through the neck opening; or the bar goes into a bottom through a leg opening or the waistband, or from the closed end, instead of into the middle fold from the open end.
- **SOP rule broken:** Steps 3.1 and 3.2, the hanger goes into a top through the neck opening and nowhere else, and the bar goes into a bottom's middle fold from the open end and nowhere else.
- **Coaching note:** neck opening for a top, open end into the fold for a bottom.

Violation: Neck stretched or hanger forced in

- **Visible cue:** the hanger presses against the neck opening going in, the neck opening pulls out of shape, a seam goes tight, the collar drags along with the hanger, or the bar is pushed against the cloth so the fold drags along, instead of the hanger being lifted out and brought in again.

- **SOP rule broken:** Steps 3.1 and 3.2, the hanger goes in with clear space around it and is never pressed against the cloth. A hanger that does not go in clean is lifted out and brought in again.
- **Coaching note:** if it does not go straight in, take it out and bring it in again. Never push against the cloth.

Violation: Top carried with a shoulder off the hanger

- **Visible cue:** the right gripper carries a top toward the rail while a shoulder end is in a sleeve, outside the top, or not in its shoulder; the left gripper lets go before the left side of the hanger is in the left shoulder; or the right shoulder is never put onto the right side of the hanger.
- **SOP rule broken:** Step 3.1, the left side of the hanger goes into the left shoulder and the right shoulder is put onto the right side, and both are on before the top leaves the fit spot.
- **Coaching note:** both shoulders on the hanger before you go to the rail.

Violation: Top hung with cloth folded wrong

- **Visible cue:** a top goes to the rail with its collar or neckband turned under, a sleeve caught up over the hanger or the hook, or the front caught up inside the hanger, and the episode moves on.
- **SOP rule broken:** Step 3.1, the collar or neckband lies flat around the hook and nothing is caught up on the hanger before the top leaves the fit spot.
- **Coaching note:** look at the collar and the sleeves before you carry. Lift anything caught and lay it flat.

Violation: Bottom off the middle of the bar

- **Visible cue:** a bottom goes to the rail with its fold near one end of the bar, one half hanging much longer than the other, the legs apart, or the fold lying crooked across the bar, and the episode moves on.
- **SOP rule broken:** Step 3.2, the fold lies across the middle of the bar, flat and level, legs together, the two halves the same length, before the bottom leaves the fit spot.
- **Coaching note:** fold in the middle, halves even. If it is off, lift the fold and lay it back.

Violation: Fold let go before it lies on the bar

- **Visible cue:** the left gripper opens before the bar is under the middle of the fold, so the bottom drops onto the bar or off it; or it opens above the bar instead of bringing the fold down onto it first.
- **SOP rule broken:** Step 3.2, the bar slides under the middle of the fold first, then the left gripper brings the fold down onto the bar, then it opens.
- **Coaching note:** bar all the way under, fold down onto it, then let go.

Violation: Garment carried the wrong way

- **Visible cue:** a top is carried to the rail by the hook alone, or by the left gripper; a bottom is carried by its cloth instead of the hook; the garment swings, turns, or is shaken on the way; or it is carried over a garment already hanging or dragged across the pile.
- **SOP rule broken:** Step 4, the right gripper carries a top level by its hold on the right shoulder and a bottom level by the hook, clear of the pile and of every garment already hanging.
- **Coaching note:** right shoulder for a top, hook for a bottom, level and around, never over.

Violation: Hanger let go before it rests on the rail

- **Visible cue:** the right gripper opens above the rail and drops the garment and hanger onto it; it lets go before the hook rests on the rail, so the garment drops the last part of the way or swings off; or a gripper opens anywhere other than at a let-go point.
- **SOP rule broken:** Step 4, the right gripper opens only once the hook rests on the rail, and a gripper opens nowhere but at a let-go point.
- **Coaching note:** all the way down, then let go. Watch it as the gripper comes off.

Violation: Hanger not fully on the rail, or out of the run

- **Visible cue:** the hook rests on top of the rail rather than over it, hangs on the hook of the hanger beside it, hangs on an upright, or sits half on with the garment tipping, and the next garment comes off the pile without this one being lifted off and hung again; or the first garment is hung somewhere other than the left end, or a garment is hung to the left of one already hanging, and the episode moves on.
- **SOP rule broken:** Step 4, the hook is brought down until it rests over the rail with the garment hanging free, the first garment goes at the left end and each one after it to the right of the last, and a bad hang is fixed by lifting the garment off by its hook and hanging it again.
- **Coaching note:** seat the hook over the bar and watch it hold before you let go. Left end first, then rightward. If it is not right, lift it off and hang it again.

Violation: Garment left slipping on the hanger

- **Visible cue:** a hung top has a shoulder off its hanger end or is sliding off the hanger, or a hung bottom has slid along the bar or is sliding off it, and the next garment comes off the pile without it being set right.
- **SOP rule broken:** Step 4, a garment hangs settled, a top with both shoulders on their hanger ends and a bottom across the middle of the bar, before the next garment comes off the pile.
- **Coaching note:** look at it after the gripper comes off. If it is slipping, set it right now.

Violation: Garment hung facing the wrong way

- **Visible cue:** a top hangs on the rail with its back toward the front edge, or a hanger is turned on the rail after it is hung.
- **SOP rule broken:** Steps 1.2 and 4, a top is held and hung with its front facing the front edge.
- **Coaching note:** front to the front edge from the lift. If it hangs the wrong way at the fit spot, put it back and take it again.

Violation: Hooks not all facing the same way

- **Visible cue:** a hook is set on the rail with its opening facing the front while the others face the back.
- **SOP rule broken:** Step 4, every hanger is set down on the rail with its hook opening facing the back.
- **Coaching note:** openings to the back, every hanger, as it goes on the rail.

Violation: Hung garment pushed or slid along the rail

- **Visible cue:** a gripper pushes, slides, or re-spaces a garment already hanging on the rail instead of lifting it off by its hook and hanging it again, or a gripper reaches through the run to get at the rail.
- **SOP rule broken:** Step 4, a bad hang is fixed by lifting the garment off the rail by its hook and hanging it again.
- **Coaching note:** lift it off and hang it again. Nothing gets pushed along the rail.

Violation: Rack moved or lifted

- **Visible cue:** the rack rocks, slides, tips, or lifts off the tabletop; the rail bends or twists under a hook; or either gripper holds, drags, leans on, or turns the rack.
- **SOP rule broken:** Step 4, the rack stands on its own base and never moves, and no gripper touches it.
- **Coaching note:** the rack stands on its own. If it is moving, stop and set it back.

Violation: Arms swapped roles or crowded the same spot

- **Visible cue:** the right gripper takes a garment off the pile or holds it for the fit; the left gripper takes a hanger or carries a garment to the rail; the right gripper pushes the left shoulder of a top onto the hanger, or the left gripper puts the right shoulder on; or the two grippers work the same spot, touch each other, or one has to move before the other can reach.
- **SOP rule broken:** Steps 1 to 4, the left gripper lifts and holds each garment, puts the left shoulder of a top on the hanger, and lays the fold of a bottom on the bar, and the right gripper takes the hanger, puts it on, puts the right shoulder of a top on, and hangs every garment.

- **Coaching note:** left lifts and holds, right fits and hangs. Neither one does the other's work.

Violation: Item dropped or something knocked over

- **Visible cue:** a garment or a hanger falls onto the tabletop or the floor; a garment swings off the rail or is knocked off it; a gripper strikes the rack or an upright and it shifts; the garment pile or the hanger pile is knocked over or dragged out of place; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 4, each item stays on a clear path, a garment already hung is left hanging on the rail, and a gripper opens only at a let-go point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with fewer than four garments on the rail, a garment on an upright, a garment slipping on its hanger, or something still on the garment pile or the hanger pile; or either arm is away from home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 5, look once along the rail, return both arms home with grippers open, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Bad garment:** a torn garment or open seam, or a top whose neck opening will not open wide enough for a hanger to pass with clear space around it. Replace it before the next episode.
- **Bad hanger:** a bent or broken hook, a cracked shoulder end, a bar whose open end has closed up, or a hanger that will not sit straight on the rail. Replace it before the next episode.
- **Rack fault:** the rack will not stand straight, an upright works loose, or the rail bends or comes out of its uprights under a hung garment. Repair or replace it before the next episode.
- **Rail too short:** all four hangers will not fit on the rail in one run.
- **Rail at the wrong height:** a hung garment touches the tabletop, or a gripper cannot reach the top of a hanging garment.

Annotation subtasks (from SOP)

1. Lift one garment off the pile and hold it at the fit spot
2. Take one hanger from the hanger pile
3. Push the hanger down through the neck opening of a top
4. Push the left shoulder onto the hanger
5. Put the right shoulder onto the hanger
6. Slide the bar into the middle fold of a bottom
7. Lay the fold on the bar
8. Hang one garment on the rail to the right of the last one
9. Return both arms home and end the episode